

ML Assignment -2

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1. Assignment Overview

This document is a complete summary of submission of Assignment-2 of Machine Learning (**AIML CZG565**). Primary objective of this project is to develop and evaluate various machine learning classification models to accurately predict whether a client will subscribe to a term deposit, based on the data from a direct marketing campaign conducted by a Portuguese banking institution. This prediction will enable the bank to optimize its marketing strategies, reduce campaign costs, and improve the overall efficiency of customer outreach.

2. GitHub Repository Details:

GitHub Repository Link:

<https://github.com/vaibhavkhare1206/ML-Assignment-Repo-2025ab05182>

This Repository Contains:

- **data/** folder contains the dataset (Train and test data) required to train, test the model and run the Application.
- **model/** folder A Complete source code for model file which have Data Analysis, Model training, Evaluation and comparison of all 6 models (In master File). Inside Model folder we have **artifacts/** and **submission_csv/** folders which contains all the model artifacts and submission files respectively
- **README.md** file which will describe entire project structure, methodology and results for **ML Classification Model Comparison for Portuguese Bank Data**
- **app.py** file for Streamlit Live Application
- **requirements.txt** file listing all required Python dependencies

3. Live Streamlit Cloud Community App

Streamlit App link:

<https://ml-assignment-repo-2025ab05182-hu4prcvy3yueihh47zzxz.streamlit.app/>

Application is developed using **Streamlit Community Cloud**. It provides the interactive User Interface for **Portuguese Bank Data**

4. How to use Streamlit App

- Click the Streamlit app link as given above, below screen will appear:

ml-assignment-repo-2025ab05182-hu4prcvy3yueihh47zzxz.streamlit.app

Configuration

Model Selection

Choose a Classification Model

Logistic Regression

You selected the Logistic Regression model.

Clear Results and Reset

Direct Marketing Campaigns on Portuguese Banking Data

This Application evaluates various machine learning models to accurately predict whether a client will subscribe to a term deposit, based on the data from a direct marketing campaign conducted by a Portuguese banking institution.

Developed by Vaibhav Khare - BITS ID: 2025ab05182@wilp.bits-pilani.ac.in

Data Upload - Upload Your Test Data

Upload your CSV file

Drag and drop file here
Limit 200MB per file • CSV

Browse files

Download sample test dataset if dataset is not available:

Download Sample 'bank-test.csv'

Please upload a CSV file to get started.

Data Preprocessing

Data Splitting

Model Training

Please upload data and complete preprocessing/splitting steps to train a model.

Model Evaluation - Prediction Report Generated Successfully

Manage app

01:49
15-02-2026

➤ Upload Test dataset for model training as highlighted. If test dataset is not available download it from application as highlighted:

ml-assignment-repo-2025ab05182-hu4prcvy3yueihh47zzxz.streamlit.app

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Model Selection

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Manage app

01:53
15-02-2026

- Select Machine learning Model from dropdown for training as highlighted:

The screenshot shows the Streamlit application interface for "Direct Marketing Campaigns on Portuguese Banking Data". The left sidebar contains a "Configuration" section with a "Model Selection" dropdown menu. The dropdown is open, showing a list of machine learning models: Logistic Regression, Decision Tree Classifier, K-Nearest Neighbor Classifier, Naive Bayes Classifier (Gaussian), Random Forest Model, and XGBoost Model. The "Logistic Regression" option is highlighted. The main content area displays the title "Direct Marketing Campaigns on Portuguese Banking Data" and a description: "This Application evaluates various machine learning models to accurately predict whether a client will subscribe to a term deposit, based on the data from a direct marketing campaign conducted by a Portuguese banking institution." Below this, it says "Developed by Vaibhav Khare - BITS ID: 2025ab05182@wilp.bits-pilani.ac.in". The next section is "Data Upload - Upload Your Test Data", which includes a "Upload your CSV file" button and a "Download sample test dataset if dataset is not available:" button. A message at the bottom says "Please upload data and complete preprocessing/splitting steps to train a model."

- Once Data is uploaded Application will give a brief analysis about test dataset, preprocess the same and will provide a brief Analysis of Pre-processed Data

The screenshot shows the Streamlit application interface after the data has been uploaded successfully. The left sidebar shows the "Model Selection" dropdown menu with "Logistic Regression" selected. The main content area displays a green message: "File uploaded successfully! Here's a preview of your data:". Below this is a table showing the first 5 rows of the dataset. The next section is "Data Preprocessing", which includes a message: "Target variable 'y' converted to numerical (1/0)." and a message: "One-hot encoding applied to categorical columns: job, marital, education, default, housing, loan, contact, month, poutcome." Below this is a section titled "Handling Missing Values" with a message: "No missing or infinite values found." The final section is "Preprocessed Data Preview:", which shows a table with the preprocessed data. The table has 17 columns: age, balance, day, duration, campaign, pdays, previous, y, job_blue-collar, job_entrepreneur, job_housemaid, job_management, job_retired, job_self-employed, job_services, job_student, job_technician, job_unemployed, and job_unknown. The first 5 rows of the table are shown.

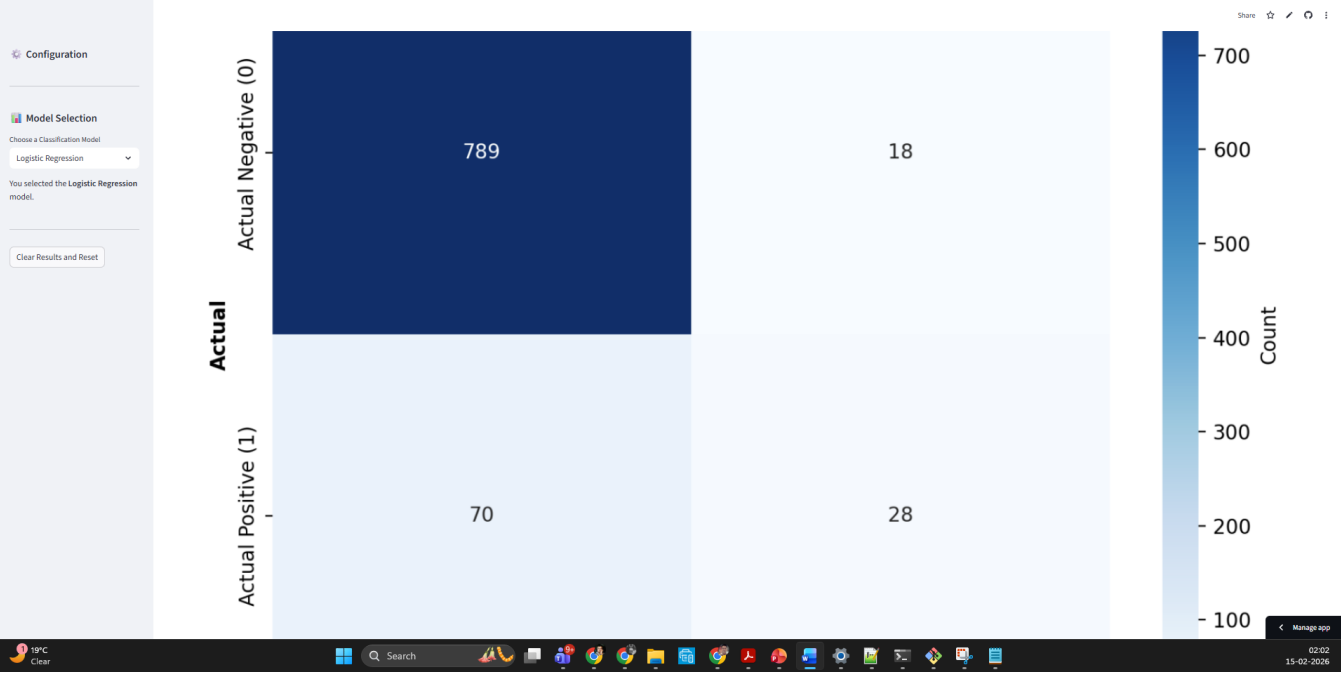
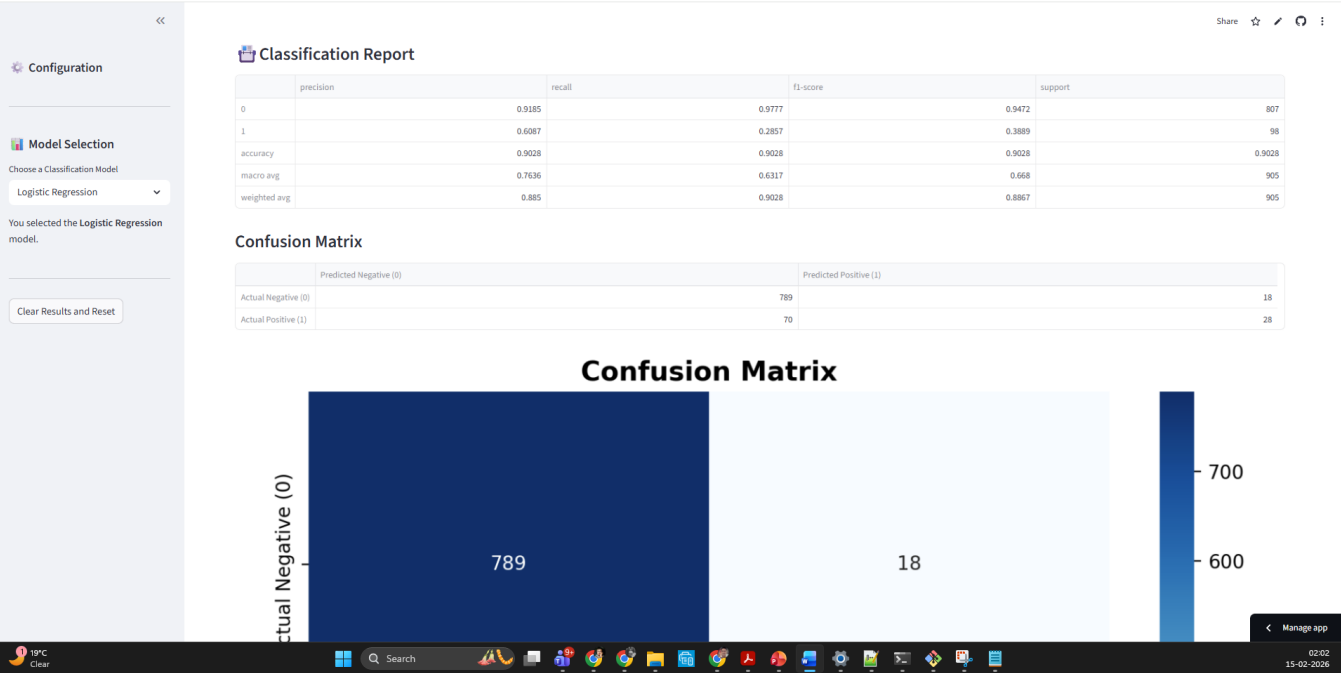
	age	job	marital	education	default	balance	housing	loan	contact	day	month	duration	campaign	pdays	previous	poutcome	y
0	30	unemployed	married	primary	no	1787	no	no	cellular	19	oct	79	1	-1	0	unknown	no
1	33	services	married	secondary	no	4789	yes	yes	cellular	11	may	220	1	339	4	failure	no
2	35	management	single	tertiary	no	1350	yes	no	cellular	16	apr	185	1	330	1	failure	no
3	30	management	married	tertiary	no	1476	yes	yes	unknown	3	jun	199	4	-1	0	unknown	no
4	59	blue-collar	married	secondary	no	0	yes	no	unknown	5	may	226	1	-1	0	unknown	no

➤ Click On Train Model as highlighted:

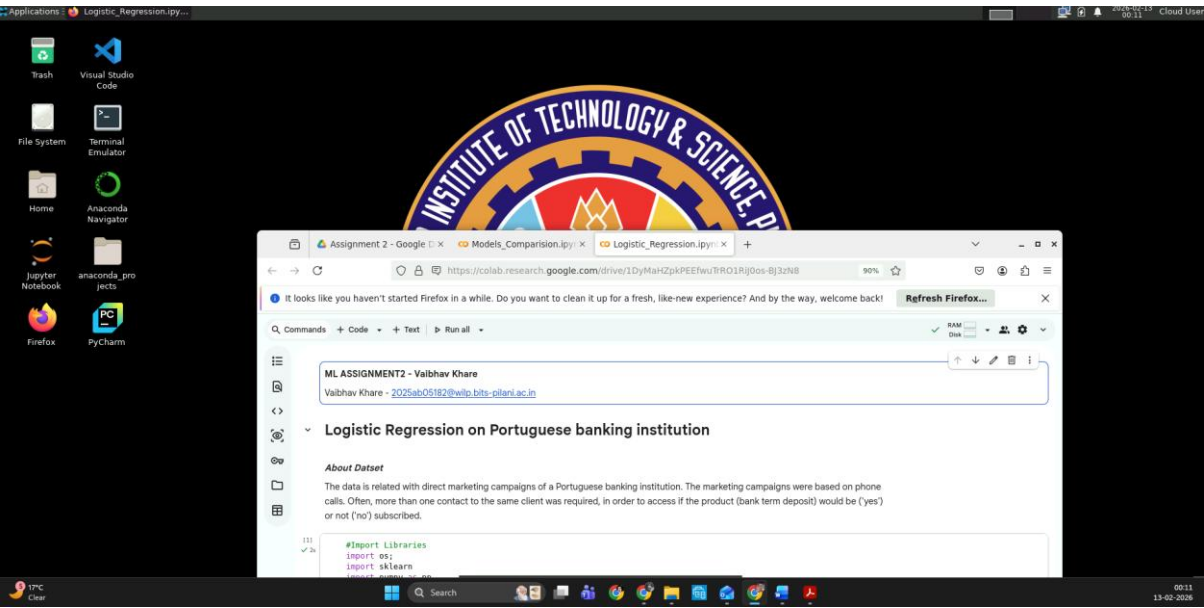
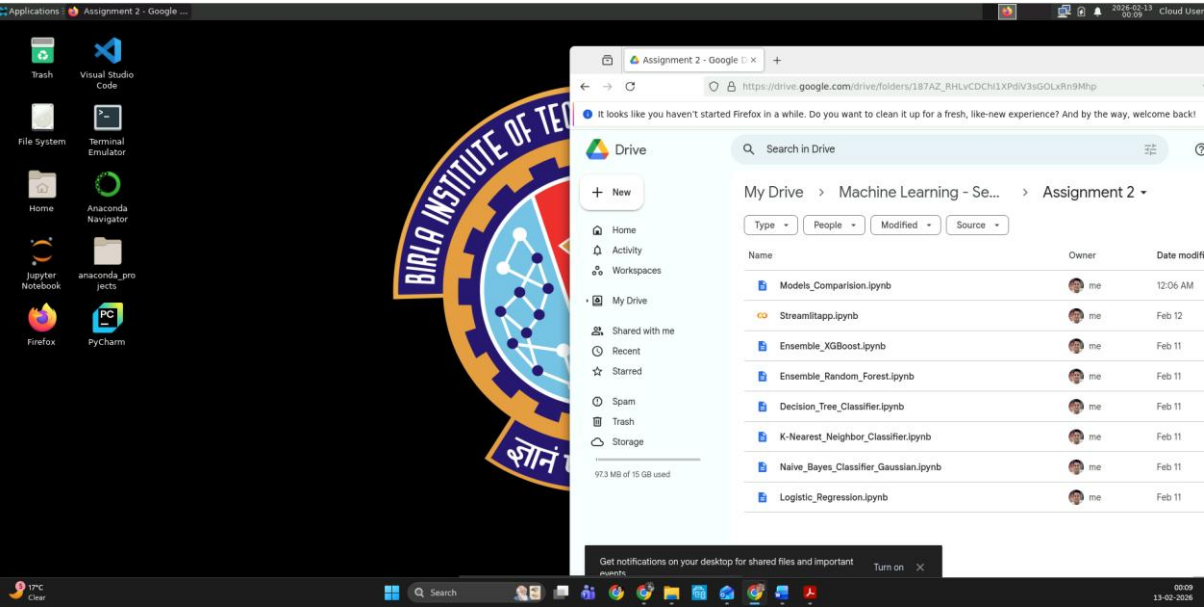
[illegible]

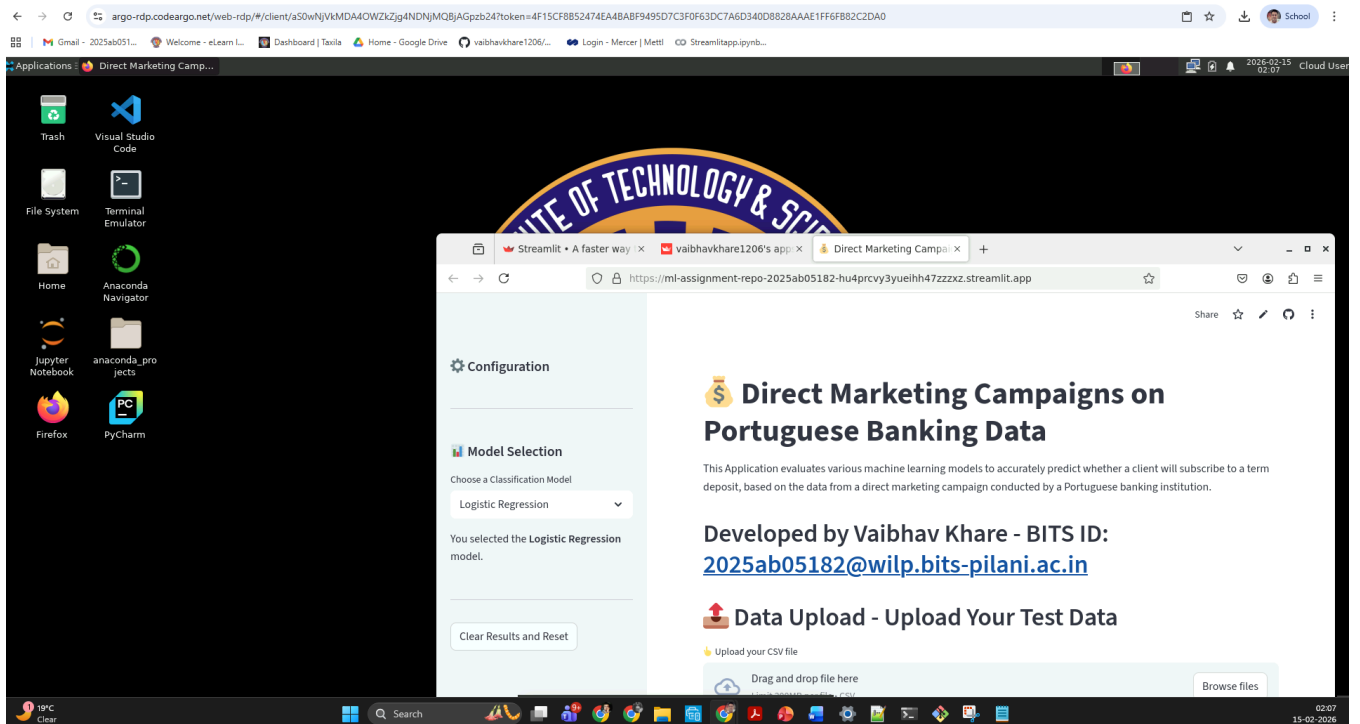
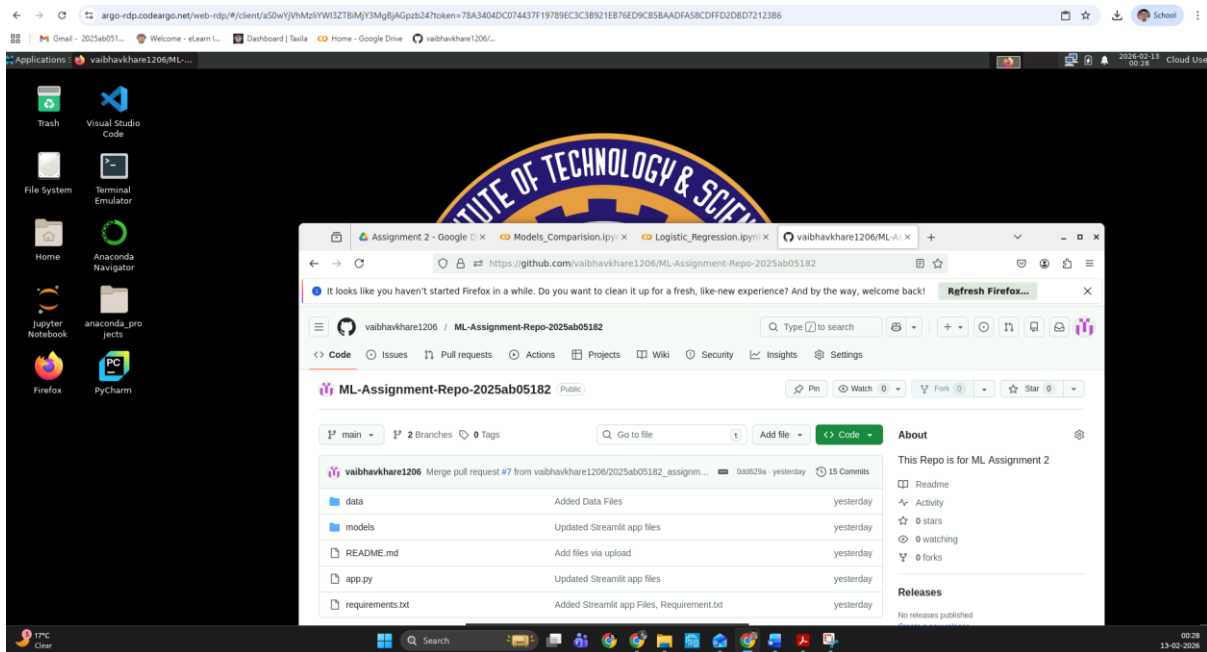
➤ Once Model is trained below results will appear:

The screenshot shows a web application interface for model training and evaluation. The browser's address bar displays the URL: `ml-assignment-repo-2025ab05182-hu4prcvy3yueihh47zzxz.streamlit.app`. The sidebar on the left contains two main sections: 'Configuration' and 'Model Selection'. Under 'Model Selection', a dropdown menu shows 'Logistic Regression' selected, with a message stating 'You selected the Logistic Regression model.' Below this is a 'Clear Results and Reset' button. The main content area is titled 'Model Training' and shows a progress bar for 'Training Logistic Regression...' which is complete. A green message box states 'Logistic Regression trained successfully!' and another green box with a checkmark says 'Model ready for evaluation.' Below this is a section titled 'Model Evaluation - Prediction Report Generated Successfully' with a green checkmark icon. Underneath, it says 'Evaluating Logistic Regression:' followed by an 'Evaluation Matrix' table. The 'Evaluation Matrix' table has two columns: 'Metric' and 'Value'. The metrics listed are Accuracy (0.9028), Precision (0.6087), Recall (0.2857), F1-Score (0.3889), AUC Score (0.8966), and Matthews C (0.3727). Below the matrix is a 'Classification Report' table. The 'Classification Report' table has five columns: 'precision', 'recall', 'f1-score', 'support', and an unlabeled column. The rows are labeled '0' and '1'. The values for row '0' are precision: 0.9185, recall: 0.9777, f1-score: 0.9472, support: 807. The values for row '1' are precision: 0.6087, recall: 0.2857, f1-score: 0.3889, support: 807. The bottom of the image shows a Windows taskbar with various icons and a system tray showing 19°C and 'Clear'.



5. BITS Virtual Lab Screenshot





Configuration

Model Selection

Choose a Classification Model

Logistic Regression

You selected the Logistic Regression model.

Clear Results and Reset

bank-test.csv 371.0KB

Download sample test dataset if dataset is not available:

Download Sample 'bank-test.csv'

File uploaded successfully! Here's a preview of your data:

	age	job	marital	education	default	balance	housing	loan	contact	day	month	duration
0	30	unemployed	married	primary	no	1787	no	no	cellular	19	oct	79
1	33	services	married	secondary	no	4789	yes	yes	cellular	11	may	220
2	35	management	single	tertiary	no	1350	yes	no	cellular	16	apr	185
3	30	management	married	tertiary	no	1476	yes	yes	unknown	3	jun	199
4	59	blue-collar	married	secondary	no	0	yes	no	unknown	5	may	226

Configuration

Model Selection

Choose a Classification Model

Logistic Regression

You selected the Logistic Regression model.

Clear Results and Reset

y_train shape: (3616,
X_test shape: (905, 42)
y_test shape: (905,)

Model Training

Train Model

Training Logistic Regression...

Logistic Regression trained successfully!

Model ready for evaluation.

6. Dataset Description:

- **Dataset:** Portuguese Bank Marketing From UCI Machine Repository
- **Source:** [UCI ML Repository - Portuguese Bank Marketing](#)

- **Description:** The data is related with direct marketing campaigns of a Portuguese banking institution. The marketing campaigns were based on phone calls. Often, more than one contact to the same client was required, in order to access if the product (bank term deposit) would be (or not) subscribed.

- **Dataset Dimensions:**

- bank_train: Contains 45211 rows and 17 columns.
- bank_test: Contains 4521 rows and 17 columns.

- **Features:**

The dataset comprises both numerical and categorical features:

- **Numerical Features:** age, balance, day, duration, campaign, pdays, previous.

Feature Name	Description
age	Age of the client.
balance	Average yearly balance of the client, in euros.
day	Last contact day of the month.
duration	Last contact duration, in seconds (highly influential, but should be removed for realistic prediction as it's known only after the call).
campaign	Number of contacts performed during this campaign and for this client.
pdays	Number of days that passed after the client was last contacted from a previous campaign (-1 means client was not previously contacted).
previous	Number of contacts performed before this campaign and for this client.

- **Categorical Features:** job, marital, education, default, housing, loan, contact, month, poutcome.

Feature Name	Description
job	Type of job (e.g., admin., blue-collar, entrepreneur, management, etc.).
marital	Marital status (e.g., married, divorced, single).
education	Level of education (e.g., primary, secondary, tertiary, unknown).
default	Has credit in default (yes/no).
housing	Has housing loan (yes/no).
loan	Has personal loan (yes/no).
contact	Contact communication type (e.g., cellular, telephone, unknown).
month	Last contact month of year (e.g., jan, feb, mar, ..., nov, dec).
poutcome	Outcome of the previous marketing campaign (e.g., failure, other, success, unknown).

- **Target Variable:** The target variable is y, which indicates whether the client subscribed to a term deposit. It originally contained yes or no values, which were subsequently mapped to 1 and 0 respectively for model training.

7. Model Evaluation Summary:

The performance of multiple machine learning models was evaluated using standard classification metrics including Accuracy, Area Under the ROC Curve (AUC), Precision, Recall, F1-score, and Matthews Correlation Coefficient (MCC). These metrics were chosen to provide a balanced assessment of predictive performance, robustness, and class-wise consistency.

The following models were evaluated: Logistic Regression, Decision Tree, K-Nearest Neighbours (kNN), Naive Bayes (Gaussian), Random Forest, and XGBoost.

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression Model	0.902	0.8966	0.8868	0.902	0.8878	0.4169
Decision Tree Classifier	0.9257	0.9205	0.9196	0.9257	0.9207	0.5972
K-Nearest Neighbours Classifier	0.9162	0.9526	0.9068	0.9162	0.9058	0.5178
Naive Bayes - Gaussian Classifier	0.8478	0.7926	0.8655	0.8478	0.8557	0.3371
Ensemble Model - Random Forest	0.9573	0.9793	0.9578	0.9573	0.9536	0.7741
Ensemble Model - XGBoost	0.9549	0.9787	0.953	0.9549	0.9533	0.7665

8. Observation and Analysis:

This section synthesizes the insights derived from the model performance metrics, training times, and confusion matrices to provide a holistic understanding of each model's strengths, weaknesses, and suitability for the direct marketing campaign problem

Model Performance Observation

ML Model Name	Observation about model performance
Logistic Regression Model	As a linear model, Logistic Regression provided a reasonable baseline (Accuracy: 0.9020, F1: 0.8878) but struggled to capture the full complexity of the data compared to the non-linear models.
Decision Tree Classifier	While better than simpler models, its performance (Accuracy: 0.9257, F1: 0.9207) indicates it captures some non-linear patterns but is surpassed by the more complex ensemble methods.
K-Nearest Neighbours Classifier	K-Nearest Neighbours showed a good AUC (0.9526) but slightly lower Accuracy (0.9162) and F1 (0.9058) compared to Decision Tree, suggesting it might be sensitive to the feature space or the chosen number of neighbours.
Naive Bayes - Gaussian Classifier	Naive Bayes was the weakest performer among all (Accuracy: 0.8478, F1: 0.8557), suggesting that the assumption of conditional independence between features given the class might not hold strongly for this dataset.
Ensemble Model - Random Forest	Random Forest model emerged as the best performer, achieving the highest accuracy (0.9573), AUC (0.9793), Precision (0.9578), Recall (0.9573), F1-score (0.9536), and MCC (0.7741). This highlights the benefit of combining multiple models or using boosting techniques for better generalization and accuracy.
Ensemble Model - XGBoost	XGBoost achieved the second-best performance accuracy (0.9549), AUC(0.9787), Precision (0.953), Recall (0.9549), F1-score (0.9533), and MCC (0.7665). It effectively captures complex patterns and feature interactions. While slightly lower than Random Forest in this case, it shows strong performance with good generalization. The model benefits from gradient boosting's ability to correct errors iteratively.

Overall Insights:

➤ Logistic Regression

- **Performance:** Moderate. Achieved an Accuracy of 0.9020 and an AUC of 0.897. Its F1 score (0.8868) and MCC (0.417) were lower than tree-based and ensemble methods, suggesting it struggles with the non-linear nature of the data.
- **Training Time:** Moderate to Slow (5.02 seconds). It was slower than Decision Tree and XGBoost, likely due to iterative optimization on a dataset with many features.

Implications: Logistic Regression has a relatively high number of False Negatives (348). This means it misses a significant portion of potential customers, which is undesirable for a direct marketing campaign where identifying potential subscribers (minority class) is crucial. While its False Positives are low, the high FN rate is a major drawback.

➤ Decision Tree Classifier

- **Performance:** Good. With an Accuracy of 0.9257 and an AUC of 0.9205, it performs significantly better than Logistic Regression and Naive Bayes. Its F1 (0.9207) and MCC (0.5972) are also solid.
- **Training Time:** Fast (1.12 seconds). It trained quickly, making it a good candidate for rapid prototyping.

Implications: The Decision Tree significantly reduces False Negatives compared to Logistic Regression (236 vs 348) while maintaining a low FP rate. This is a positive step towards identifying more potential customers, showing better sensitivity to the minority class.

➤ K-Nearest Neighbours (KNN) Classifier

- **Performance:** Moderate. It achieved good Accuracy (0.9162) and a strong AUC (0.9526), indicating good discrimination. However, its F1 (0.9058) and MCC (0.5178) were lower than the Decision Tree.
- **Training Time:** Extremely Fast (0.03 seconds). This is typical for instance-based learning where most computation occurs during prediction.

Implications: KNN has a higher False Negative count (302) than the Decision Tree and significantly more than the top ensemble models. While its FP count is low, its ability to correctly identify the minority class is not as strong as tree-based models, despite a high AUC.

➤ Naive Bayes - Gaussian Classifier

- **Performance:** Lowest. It consistently showed the lowest performance across all metrics (Accuracy: 0.8478, AUC: 0.7926, F1: 0.8557, MCC: 0.3371).
- **Training Time:** Extremely Fast (0.06 seconds). Its simplicity allows for very quick training.

Implications: Naive Bayes exhibits a very high False Positive rate (416), meaning it frequently misidentifies non-subscribers as potential customers, leading to wasted marketing efforts. It also has a considerable number of False Negatives (272). This model is not suitable for this problem due to its poor overall performance and high FP count.

➤ Ensemble Model - Random Forest

- **Performance: Best Model** This model is the top performer, achieving the highest scores in Accuracy (0.9573), AUC (0.9793), Precision (0.9578), Recall (0.9573), F1 (0.9536), and MCC (0.7741).
- **Training Time:** Slowest (8.86 seconds). Its superior performance comes at the cost of the longest training time among all models.

Implications: Random Forest excels at minimizing both False Positives (only 11) and False Negatives (182). This means it is very precise in identifying actual non-subscribers and effectively captures a large proportion of true potential customers. The low FN count is particularly beneficial for a marketing campaign.

➤ Ensemble Model - XGBoost

- **Performance:** Second Best. XGBoost is a very close second to Random Forest, with excellent scores across the board (Accuracy: 0.9549, AUC: 0.9787, F1: 0.9533, MCC: 0.7665).
- **Training Time:** Fast (0.86 seconds). Remarkably, XGBoost offers near-best performance while being one of the fastest models to train, significantly quicker than Random Forest.

Implications: XGBoost has a very low False Negative count (141), which is even better than Random Forest in this regard, meaning it's highly effective at identifying potential customers. Its False Positive count (63) is higher than Random Forest but still very low, indicating good overall classification.

9. Gitlab Repository Structure:

```
portuguese-bank-data-classification
|
|— data/                # Dataset files
|   |— bank-full.csv
|   |— bank-names.txt
|   |— bank-test.csv
|— model/              # Model directory
|   |— artifacts/      # Saved models pkl files
|       |— all_confusion_matrices.png
|       |— decision_tree.pkl
|       |— knn.pkl
|       |— logistic_regression.pkl
|       |— model_comparison_visualization.png
|       |— naive_bayes.pkl
|       |— random_forest.pkl
|       |— scaler.pkl
|       |— xgboost.pkl
|   |
|   |— submission_csv/  # Training results
|       |— decision_tree_classifier_submission.csv
|       |— k-nearest_neighbour_submission.csv
|       |— logistic_regression_submission.csv
|       |— model_comparison.csv
|       |— naive_bayes_submission.csv
|       |— random_forest_submission.csv
|       |— xgboost_submission.csv
|   |— Decision_Tree_Classifier.ipynb # Training notebooks
|   |— Ensemble_Random_Forest.ipynb
|   |— Ensemble_XGBoost.ipynb
|   |— K-Nearest_Neighbor_Classifier.ipynb
|   |— Logistic_Regression.ipynb
```

```
| |— Models_Comparision.ipynb
| |— Naive_Bayes_Classifier_Gaussian.ipynb
| |— Streamlitapp.ipynb
|
|— README.md          # This file
|— app.py              # Streamlit web application
|— requirements.txt    # Python dependencies
```

10. Conclusion:

- **Ensemble Models (Random Forest, XGBoost):** These models demonstrate superior performance, achieving the highest scores across all metrics. Their ability to combine multiple decision trees helps in reducing variance and improving predictive accuracy, making them robust choices for this dataset. This suggests that the underlying relationships in the data are complex and benefit from the non-linear, high-capacity learning of these ensemble methods.
- **Tree-based Models (Decision Tree):** The Decision Tree performs reasonably well, outperforming simpler models like Logistic Regression and Naive Bayes. It shows a good balance of precision and recall. However, it's outperformed by ensemble methods, indicating that while individual trees are effective, combining them yields better generalization.
- **K-Nearest Neighbours (KNN):** KNN shows competitive AUC scores, suggesting good discrimination ability, but its other metrics (Accuracy, Precision, Recall, F1, MCC) are not as high as the ensemble methods. It falls in the middle range of performance. Its high AUC relative to its accuracy might imply a decent ability to rank positive instances, even if the overall classification boundary isn't as precise.
- **Linear and Probabilistic Models (Logistic Regression, Naive Bayes):** These simpler models perform significantly worse than the tree-based and ensemble models. This indicates that the relationships between features and the target variable (y) are likely non-linear and not well-captured by linear boundaries or simple probabilistic assumptions. Naive Bayes, in particular, suffers from its strong assumption of feature independence, which is rarely true in real-world datasets, leading to the lowest performance.